Mahmoud Saeed Mansour

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Education & Military

Pharos University 2022 – 2026
Alexandria, Egypt
B.Sc. in Computer Science & AI
GPA: 3.76

Military Status 2026
Final Exemption

Experience

iSchool – *Tutor* Jul – Aug 2026
Remote, Egypt
Taught Python, AI, Data Science, and Godot; guided students in building and debugging projects.

Training

NTI Aug – Sep 2025
Natural Language Processing
Online

European Educational Group
Deep Learning Jul – Sep 2024
Alexandria, Egypt

Amit Jan – Aug 2024
Machine Learning & AI Diploma
Alexandria, Egypt

New Horizons Sep 2023 – Jan 2024
Flutter Application Development
Alexandria, Egypt

Technical Skills

Languages: Python, Java, Dart, C#
Software: Flutter, PyQt5, Flask, TinkerCAD, Java-GUI
AI/ML & Libraries: CNN, TensorFlow, PyTorch, OpenCV, Scikit-learn, SciPy, MediaPipe, NumPy, Pandas, YOLO (Ultralytics), Excel
Concepts: AI, Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP, Data Science
Operating Systems: Windows, Linux (Ubuntu)

Languages

Arabic: Native
English: Intermediate

Projects

› **Face Mask Segmentation during COVID-19** | *Python*, *CNN*, *TensorFlow*, *Keras*, *Flask*

- Developed a binary segmentation model using ResNet50 and VGG16 with custom upsampling.
- Deployed the model using Flask with an HTML/CSS interface for real-time mask detection.

› **Real-Time Autonomous Driving Perception System** | *Python*, *YOLOv8*, *ResNet50*, *OpenCV*, *Kalman Filter*

- Developed real-time perception combining YOLOv8 object detection with hybrid lane tracking (ResNet50, Kalman Filter, Hough Transform).
- Built a threaded Tkinter GUI with dual live streams, real-time annotations, and FPS monitoring.

› **Travel Planner Chatbot** | *Python*, *Transformers*, *HuggingFace*, *FAISS*, *Streamlit*

- Developed a RAG travel chatbot using Qwen2.5 and OSU TravelPlanner dataset for personalized recommendations.
- Integrated FAISS for semantic retrieval and SpeechRecognition for voice-based queries.
- Built a Streamlit interface for multi-day itinerary generation with recommendation explanations.

› **Vehicle License Plate & Speed Detection System** | *Python*, *YOLOv8*, *OpenCV*, *DeepSORT*

- Developed real-time vehicle detection and tracking using YOLOv8 and DeepSORT.
- Used OpenAI Vision to detect license plates and extract vehicle information.
- Estimated vehicle speeds and generated annotated videos with tracking IDs and license plates.

› **Bitcoin Price Prediction Web Application** | *Python*, *PyTorch*, *LSTM*, *GRU*, *Flask*

- Developed LSTM and GRU models for Bitcoin price forecasting across 7, 15, 30, 60, and 90-day horizons.
- Implemented data preprocessing using scaling, sliding windows, and inverse transformation.
- Built a modular Flask application and REST API with automatic model loading for price predictions.

› **Bioinformatics Cancer DNA Classification** | *Python*, *ML*, *Deep Learning*, *Bioinformatics*

- Built ML/DL models to classify cancer types from DNA sequence data using bioinformatics techniques.
- Processed genomic data through encoding, feature extraction, and dimensionality reduction.
- Compared multiple models and optimized performance using classical ML and neural networks.
- Evaluated models using accuracy, precision, recall, and ROC-AUC for robust classification.

› **Gender Identification System** | *Python*, *Deep Learning*, *CNN*, *GMM*, *OpenCV*

- Built a CNN-based system for gender classification from facial images.
- Applied preprocessing techniques including resizing, normalization, and data augmentation.
- Combined CNN features with GMM for improved classification performance.
- Integrated OpenCV for face detection and real-time image/video inference.

Graduation Project 2025 – 2026

› **The Guardian – AI-Powered Smart Home Security Robot** | *YOLOv8*, *RL*, *Flutter*, *Supabase*, *Raspberry Pi 5*, *IoT*, *Zero-Trust*

- Engineered an autonomous mobile security robot on **Raspberry Pi 5**, deploying custom **YOLOv8** and **OpenCV** for low-latency human detection, face recognition, and weapon identification.
- Trained a **PPO-based** (Proximal Policy Optimization) Reinforcement Learning agent enabling autonomous navigation, dynamic obstacle avoidance, and optimized patrol routing.
- Developed a cross-platform **Flutter** mobile app for live monitoring, instant alerts, and remote manual override, backed by **Supabase** for auth, Row-Level Security (RLS), and real-time event sync.
- Architected an end-to-end **Zero Trust Architecture** (encrypted WebSocket, secure API auth, SHA-256 verification) with multi-threaded edge-AI processing across camera modules and ultrasonic sensors.